

sTMS Interface Functional Specification

Singapore Polytechnic – Pace Website

Version 1.0

Document Version

Version	Effective Date	Summary of Changes	Created/ Updated By
0.1	31/05/2022	The Initial version of SP Pace Website IFS	AvePoint
0.2	19/07/2022	Based discussion on 13 July 2022, the changes to the following: 1. Technical interface transfer to be SFTP instead of API 2. Interface Field Mapping 3. Inclusion of sample files in SFTP	AvePoint
0.3	15/09/2022	Revised based on meeting feedback.	AvePoint
0.4	19/09/2022	Revised based on meeting feedback.	AvePoint
1.0	20/09/2022	Official version for acceptance and sign-off.	AvePoint

Document Acceptance Form

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Table of Contents

Document Version	1
Document Acceptance Form	2
1. Introduction	4
1.1 Purpose of this document	4
1.2 Intended Audience	4
2. Technical Overview	5
2.1 System Flow.....	5
2.2 Interface Design Summary	6
2.3 Business Rule.....	7
3. Interface Fields	8
3.1 Common Interface Fields.....	8
3.2 About Course Fields	8
3.3 Course Structure Fields	10
3.4 Course Schedule Fields	12
3.5 Entry Requirements Fields	15
3.6 Financial Matters Fields.....	16
3.7 Additional Information Fields.....	18
4. Interface Schemas.....	19
5. Interface Fields Mapping List.....	26
6. Assumption and Constraints.....	27
6.1 Assumption.....	27
6.2 Constraints.....	27

1. Introduction

1.1 Purpose of this document

This Interface Functional Specification (IFS) is to provide the agreed baseline of communication requirements and detailed technical specifications between the SaaS shared Training Management System (sTMS) and Singapore Polytechnic Pace Website for publishing CET course and schedule information.

sTMS shall identify all new created, updates and deletions of course information to generate corresponding JSON file. sTMS will generate and deposit the JSON file encrypted by PGP protocol in SFTP server accordingly. Singapore Polytechnic PACE Website would be able to retrieve course information based on the transferred JSON file.

Students can view the relevant information of the course on the SP Pace website, and click the "Register Here" button to be redirected to sTMS to complete the registration.

1.2 Intended Audience

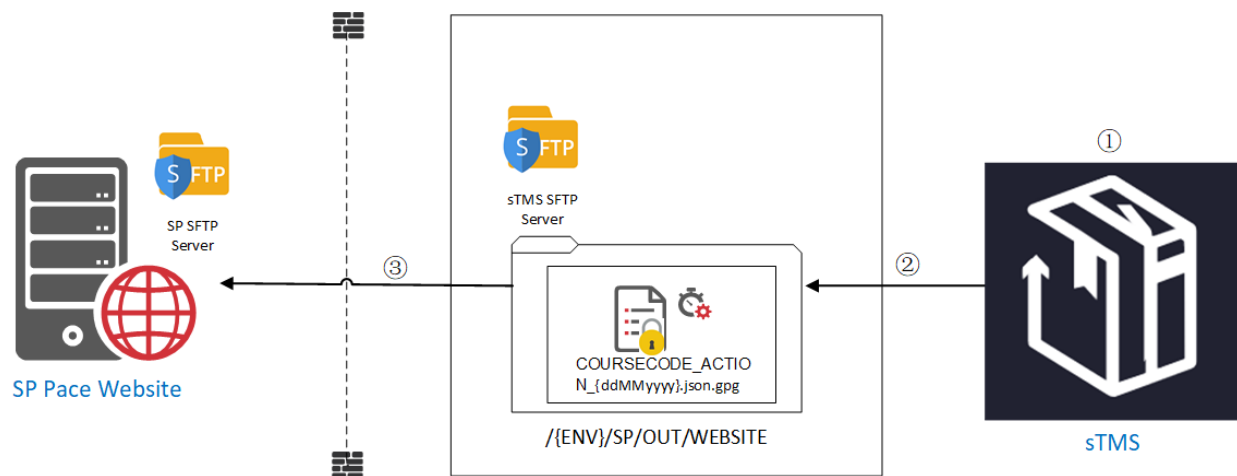
This technical document is used for system communication. Therefore, the intended audiences are the technical stakeholders of SP Pace Website and sTMS.

2. Technical Overview

2.1 System Flow

There are 3 roles involved in the interface diagram.

- **SP Pace Website:** client application that consumes the API.
- **sTMS SFTP Server:** A SFTP server hosted in Azure that stores encrypted JSON file used for data exchange.
- **sTMS:** shared Training Management System.



SN	Description
1	sTMS generates the course info JSON file based on SP course code and course catalogue status on a pre-defined schedule.
2	sTMS configures timer job to push course catalogue info JSON file into sTMS SFTP server after encryption with PGP protocol.
3	sTMS SFTP server push JSON file into SP SFTP server.

2.2 Interface Design Summary

Interface File Name	COURSECODE_ACTION_{ddMMyyyy}.json.gpg
Data Source System	shared Training Management System
Data Destination System	Singapore Polytechnic Pace Website
Mode of Interface	JSON
Method	SFTP
File Path	/{ENV}/SP/OUT/PACE ENV: UAT, PROD
Encryption Method	PGP
Frequency	Daily sync
Timing	Timer job trigger timing is configurable in sTMS.
File Retention Policy	Files are retained for a maximum of 30 days in the sTMS SFTP server, after this period the files will be automatically deleted
File Transfer Policy	sTMS shall send an empty file (Sample JSON file format is: EMPTY_{ddMMyyyy}.json) if all SP course catalogue status and its data fields don't have any updates
Possible Exception and Exception Handling	File loss or transfer failure due to sTMS internal exception. E.g. Timer service down, etc. Exception handling steps: • sTMS support team shall be informed by email of the transfer failure. • sTMS support team checks the relevant logs to troubleshoot the cause of the failure. • If the issue is due to the transfer file could not be found in SP SFTP server: ○ sTMS support team will manually rerun the timer job to trigger file generation and deposit the file in SP SFTP server.

2.3 Business Rule

Course Catalogue Info JSON File Generation Rule

After sTMS go-live, the first time synchronization sTMS will send all the SP published course catalogue information to SP Pace Website.

The subsequent data sync will be triggered when course catalogue modified time later than last job time.

The SP Pace Website interface is to sync SP course information published in the sTMS course catalogue. Editing or creating draft version of course catalogue will not trigger the interface job.

3. Interface Fields

3.1 Common Interface Fields

Name	Description	Data Type
CourseCode	Course code.	String(\$nvarchar(20))
CourseName	Course name.	String(\$nvarchar(60))
PublishTime	Course catalogue publish time.	String(\$dd-MM-yyyy hh:mm:ss)
CourseType	Course type code.	String(\$nvarchar(20))
CourseCategory	Course category code.	String(\$nvarchar(20))
BranchCode	Branch code.	String(\$nvarchar(60))
CourseDetailLink	Course detail URL for sTMS.	String(\$nvarchar(max))

3.2 About Course Fields

Name	Description	Data Type	Pace Website Fields
Image	Course image	String(\$byte)	Image
Objective	Course objective	String(\$nvarchar(max))	Course Objective
Description	Course description	String(\$nvarchar(max))	Course Objective
TargetAuds	Specifies the target audience of the course	String(\$nvarchar(max))	Suitable for
AdditionalInfo	Course additional information may contain the following information: Future Progression Career Opportunity General Intake Info (e.g. 2 intakes per year) Testimonial Trainer profile	String(\$nvarchar(max))	NA

3.3 Course Structure Fields

Name	Description	Data Type	Pace Website Fields
CertificateName	Course level certificate name	String(\$nvarchar(60))	Course Introduction
CertificateDetail	Details of course certificate	String(\$nvarchar(max))	Certificate / Accreditation
Certificates	Contains one or more certificates	Array of [CourseStructure.Certificates]	Course Introduction
CourseStructure.Certificates			Course Introduction/ Award/ Certification / Accreditation
CertificateName	Module level certificate name	String(\$nvarchar(60))	
Description	Certificate description	String(\$nvarchar(max))	
Modules	Contains one or more modules	Array of [CourseStructure.Certificate.Modules]	
CourseStructure.Certificate.Modules			
ModuleCode	Module Code	String(\$nvarchar(20))	
ModuleName	Module name	String(\$nvarchar(60))	
CertificateName	Module level certificate name	String(\$nvarchar(60))	
Description	Module description	String(\$nvarchar(max))	

Course structure

Course certificate

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Post Diploma Certificate 1

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Post Diploma Certificate 1



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[BM0001]e-Business Management	Lorem ipsum dolor sit amet, consetetur sadipscing elitr, sed diam nonumy eirmod tempor invidunt ut labore et dolore magna aliquyam ...
Digital Development with Python	Lorem ipsum dolor sit amet, consetetur sadipscing elitr, sed diam nonumy eirmod tempor invidunt ut labore et dolore magna aliquyam ...
Big data analytics and Optimisation	Lorem ipsum dolor sit amet, consetetur sadipscing elitr, sed diam nonumy eirmod tempor invidunt ut labore et dolore magna aliquyam ...

Post Diploma Certificate 2

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Post Diploma Certificate 2



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Big data analytics and Optimisation	Lorem ipsum dolor sit amet, consetetur sadipscing elitr, sed diam nonumy eirmod tempor invidunt ut labore et dolore magna aliquyam ...

Assessment type

Project

3.4 Course Schedule Fields

Name	Description	Data Type	Pace Website Fields
Description	Course schedule description	String(\$nvarchar(max))	Overview
Intakes	Contains one or more course intake	Array of [CourseSchedule.Intakes]	NA
CourseSchedule.Intakes			
ShowToApplicant	Couse intake info show to applicant or not Value list: Y – Yes; N - No	String(\$nvarchar(1))	Registration Period
RegistrationStartDate	Course registration start date	String(\$DD-MM-YYYY)	
RegistrationEndDate	Course registration end date	String(\$DD-MM-YYYY)	
CourseStartDate	Course start date	String(\$DD-MM-YYYY)	Overview - Course Date/ Date of Course Commencement
CourseEndDate	Course end date	String(\$DD-MM-YYYY)	
ScheduleInfo	Course schedule info description	String(\$nvarchar(max))	Overview - Time & Class Schedule
RegistrationPriority	Applicants fall under the below priority group can enjoy an earlier registration start date.	Array of [CourseSchedule.Intakes.RegistrationPriority]	NA
ModeOfTraining	Mode of training info	String(\$nvarchar(60))	Overview - Mode of Training
Venue	Venue info	String(\$nvarchar(max))	Overview - Venue
CatalogueLink	Course catalogue URL	String(\$nvarchar(255))	Register (link)
ClassPeriod	Module duration	Array of [CourseSchedule.Intake.ClassPeriod]	NA
CourseSchedule.Intakes.RegistrationPriority			NA
GroupType	Priority group type. E.g., On-going students; SG citizen; SG PR; etc.	String(\$nvarchar(60))	
RegistrationStartDate	Earlier registration start date	String(\$DD-MM-YYYY)	
CourseSchedule.Intake.ClassPeriod			
ModuleName	Module name	String(\$nvarchar(60))	
Duration	Module recur day and time	String(\$nvarchar(255))	

Course schedule

Course Schedule

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Course intake

Course intake

Course duration

Email

Schedule info

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Post Diploma Certificate 1

[BM0001]e-Business Management	Monday 10:00-14:00
Digital Development with Python	Friday 10:00-14:00
Big data analytics and Optimisation	Friday 14:00-18:00

Post Diploma Certificate 2

[BM0001]e-Business Management	Monday 10:00-14:00
Digital Development with Python	Friday 10:00-14:00
Big data analytics and Optimisation	Friday 14:00-18:00

[View class period](#) 

11/06/2021 - 01/07/2021

Ashley Lim

+6512465678

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Group type	Registration start date
On-going students	12/01/2021
SG citizen	23/02/2021
SG PR	25/03/2021
SG citizen, on-going student	27/03/2021
SG PR, on-going student	12/04/2021
Alumni	21/05/2021

 The intake registration is currently full, new application will be put under waitlist.

Apply

Course intake 2

[View class period](#)

Course intake No.	202101001	Registration duration	11/06/2021 - 01/07/2021
Course duration	24/10/2021 - 31/10/2022	Contact person	Ashley Lim
Email	Ashley@email.com	Phone	+6512465678

Schedule info

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⚠ The intake is full, no more application is accepted. Interested applicants can register interest for the course, we will notify you when there is new intake open for registration

[Apply](#)
[Interested](#)

Course intake 3

[View class period](#)

Course intake No.	202101001	Registration duration	11/06/2021 - 01/07/2021
Course duration	24/10/2021 - 31/10/2022	Contact person	Ashley Lim
Email	Ashley@email.com	Phone	+6512465678

Schedule info

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[Apply](#)

3.5 Entry Requirements Fields

Name	Description	Data Type	Pace Website Fields
EntryRequirementRemarks	Entry requirement remarks	String(\$nvarchar(max))	Minimum Entry Requirement / Assumed Skills and Knowledge

Entry requirements

Condition 1

Highest qualification

Post-Secondary; GCE A Level; H1 Gujarati; Minimum grade: A; Graduate within 5 years.

And

Certification

Big data analysis

And

Work experience

Full time; Fabricated Metal Products, Machinery & Equipment; 3 years

Or

Condition 2

Highest qualification

Post-Secondary; GCE A Level; H1 Gujarati; Minimum grade: A; Graduate within 5 years.

And

Work experience

Part time; Electronic, Computer & Optical Products; 5 years

And

Color blindness

Pass

Remarks

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3.6 Financial Matters Fields

Name	Description	Data Type	Pace Website Fields
FeePayables	Course fee payable info	Array of [FinancialMatter.FeePayables]	Fee & Funding
GST	GST remarks to informs the applicate. <i>Example:</i> The course fees payable above are inclusive of 9% GST rate.	String(\$nvarchar(max))	Fee & Funding - GST Description
PaymentOption	Course payment option.	String(\$nvarchar(max))	NA
AllowPaymentBy	Details of eligible grants and payment methods available for the course. <i>Example:</i> SkillsFeature Credit; Credit/Debit card.	String(\$nvarchar(60))	Application Procedure - Payment mode
OtherFee	Any other supplementary fees not included in course fees	String(\$nvarchar(max))	Other Fees Payable
Policy	Details of refund and withdrawal policy	String(\$nvarchar(max))	Fee & Funding
FundingScheme	Details of the funding schemes will be shown here, if any funding scheme is available	String(\$nvarchar(max))	Funding Incentive
FinancialMatter.FeePayables			Fee & Funding
FeeGroupType	Fee payable group type. E.g: MES (Singapore Citizen aged 40 years and above); SC; SPR; SC SME; SPR SME	String(\$nvarchar(max))	
Fees	Fee details	Object	
FinancialMatter.FeePayables.Fees			
CourseFeeType	Course fee type. E.g., PDC1; PDC2; Full course fee	String(\$nvarchar(max))	
Amount	Fee amount	Number	

Financial matters

Course fee payable

	PDC1	PDC2	Total course fee
Full course fee	S\$2,099.34	S\$4,381.65	S\$6,480.99
Singapore Citizen aged 40 years and above	S\$209.93	S\$438.17	S\$648.10
Singapore Citizen	S\$314.90	S\$657.25	S\$972.15
Singapore Permanent Resident	S\$839.74	S\$1,752.66	S\$2,592.40
Singapore Citizen sponsored by SME	S\$216.80	S\$452.50	S\$669.30
Singapore Permanent Resident sponsored by SME	S\$251.14	S\$524.16	S\$775.30

GST

Please be noted that course fee are subject to 7% GST.

Payment mode

Payment needs to be completed before course start on each semester

Allow payment by

SkillsFuture Credit; Credit/Debit card.

Nett supplementary fees payable

- Lab fee \$566.07, subject to GST, fee payable \$605.70, Allow payment by Skillsfuture Credit, Credit Card. (for foreigners)
- Student Union fee \$100.00, not subject to GST, fee payable \$100.00, Allow payment by Credit Card.

Refund and withdraw policy

[View details](#)

Know more about the funding scheme

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3.7 Additional Information Fields

Name	Description	Data Type	Pace Website Fields
Video	Course video link	String(\$nvarchar(255))	Video
VideoTitle	Course video title	String(\$nvarchar(255))	Video title
AreaOfInterest	Course area of interest	String(\$nvarchar(max))	Topic

About course

Course structure

Course schedule

Entry requirements

Financial matters

Area of interest

Business

Accounting

Business Law

Event Management

Watch the video

Welcome on board!

Course objectives

Fusing core concepts of engineering, finance, function, aesthetics and lifestyle, the Diploma in Engineering Design with Business (DEDB) equips you with the know-how to transform your entrepreneurial dreams into a viable business plan.

Through DEDB, you will gain essential skills and knowledge in both business and engineering aspects that will give you a head start to launch your start-up ideas in various industries. You will be exposed to practical learning experience and the opportunity to further hone your creativity when you undergo internship at established companies.

DEDB's multidisciplinary perspective will prepare you well to be an integral part of Singapore's new innovation driven economy.

Course description

The Diploma in Applied Science (Environmental Services and Management), or PTD ESM, provides opportunities for fresh ITE graduates to deepen their skills and knowledge needed for a career in the environmental services related industries.

Participants enrolled in this programme will be equipped with the necessary knowledge and skills to manage the environmental operations through adoption of technology to improve productivity.

Target audiences

Singapore citizen

Future Progression

Sneader also identified the following eight forces, which are expected to reshape the "next normal," or post-

4. Interface Schemas

Interface Schemas are detailly described in the following table:

CourseInfo ▾ {		←
Action	string	
	Action method of the course catalogue information	
CourseCode	string	
	Course code	
CourseName	string	
	Course name	
CourseType	string	
	Course type	
CourseCategory	string	
	Course category	
ModifiedTime	string(\$DD-MM-YYYY hh:mm:ss)	
	Course catalogue modified time	
CourseStatus	string	
	Course status	
CatalogueStatus	string	
	Course catalogue status	
About	AboutInfo > {...} ←	
Structure	CourseStructure > {...} ←	
Schedules	> [...]	
Requirement	EntryRequirement > {...} ←	
FinancialMatter	FinancialMatterInfo > {...} ←	
Other	OtherInfo > {...} ←	
}		

AboutInfo ▾ {

Image	string(\$byte)
	Course image
Objective	string
	Course objective
Description	string
	Course description
TargetAuds	string
	Specifies the target audience of the course
AdditionalInfo	string
	Course additional information

}

CourseStructure ▾ {

CertificateName	string
	Name of course level certificate
CertificateDetail	string
	Details of course certificate
AssessmentType	string
	Type of assessment that students will be expected to complete for the course
Certificates	▾ [Certificate > {...}]

}

```

Certificate ▾ {
  CertificateName      string
                        Module level certificate name

  Description          string
                        Certificate description

  Modules              ▾ [Module > {...} ↵]

}

```

```

Module ▾ {
  ModuleCode          string
                        Module Code

  ModuleName          string
                        Module name

  Description          string
                        Module description

}

```

```

CourseSchedule ▾ {
  Description          string
                        Course schedule description

  Intakes              ▾ [
                        Contains one or more course intake

                        CourseIntake > {...} ↵]

}

```

```

CourseIntake ▾ {
  ShowToApplicant      string
                        Couse intake info show to applicant or not.

  RegistrationStartDate string($DD-MM-YYYY)
                        Course registration start date

  RegistrationEndDate   string($DD-MM-YYYY)
                        Course registration end date

  CourseStartDate       string($DD-MM-YYYY)
                        Course start date

  CourseEndDate         string($DD-MM-YYYY)
                        Course end date

  ScheduleInfo          string
                        Course schedule info

  RegistrationPriority   ▾ [RegistrationPriority > {...} ↵]
  ModeOfTraining        string
                        Mode of training info

  Venue                string
                        Venue info

  CatalogueLink         string
                        Course catalogue URL

  ClassPeriod           ▾ [ClassPeriod > {...} ↵]
}

```

RegistrationPriority ▾ {

GroupType string
 Priority group type

RegistrationStartDate string(\$DD-MM-YYYY)
 Earlier registration start date

}

ClassPeriod ▾ {

ModuleName string
 Module name

Duration string
 Module recur day and time

}

EntryRequirement ▾ {

EntryRequirements string
 Entry requirement description

}


```

FinancialMatterInfo {
  FeePayables {
    Course fee payable info

    FeePayable {
      GST string
      Informs the applicant on whether GST is inclusive or exclusive

      AllowPaymentBy string
      Details of eligible grants and payment methods available for the course

      PaymentMode string
      Description of the deadline for completing the payment

      OtherFee string
      Any other supplementary fees not included in course fees

      Policy string
      Details of refund and withdrawal policy

      FundingScheme string
      Details of the funding schemes will be shown here, if any funding scheme is available
    }
  }
}

```

```

FeePayable {
  CertificateFees {
    FeeInfo {
      FeeGroupType string
      Course fee payable group type
    }
  }
}

```

FeelInfo ▾ {

CourseFeeType	string
	Course fee type
Amount	number
	Fee amount

}

OtherInfo ▾ {

Video	string
	Course video link
VideoTitle	string
	Course video title

}

sample JSON files:



SP-SD-FQ0001_C_29
0620221601.json



SP-SD-FQ0002_U_13
072022.json



SP-SD-FQ0003_D_13
072022.json



EMPTY_14072023.jso
n

5. Interface Fields Mapping List

The mapping relationship between sTMS course catalog fields and SP Pace Website fields are described in the following table:

sTMS Fields		SP Pace Fields
About	Image	Image
	Course Objectives	Course Objective
	Course Description	Course Introduction
	Target Audiences	Suitable for
	Course additional information (Future Progression & Trainer Profile & etc.)	NA
Course Structure	Course Certificate	Certification / Accreditation
	Module Name & Description	Course Outline
Course Schedule	Course intake & Schedule info & Apply	Course Date/ Date of Course Commencement
		Registration Period
		Time & Class Schedule
		Mode of Training
		Venue
Entry Requirements	Minimum Requirements	Register (link)
		Minimum Entry Requirement / Assumed Skills and Knowledge
Financial Matters	Course Fee Payable	Fee & Funding
	GST	Fee & Funding
	Payment Mode	Register, Application Procedure
	Allow payment by	Application Procedure
	Nett Supplementary fees payable	Other Fees Payable
	Refund and withdrawal policy	Fee & funding
	Know more about funding scheme	Fee & Funding
Other Information	Video	Video
	Video Title	Video Title
	Area Of Interest	Topic

6. Assumption and Constraints

6.1 Assumption

No.	Assumptions

6.2 Constraints

No.	Constraints
1	Illegal characters appearing in course code of JSON filename will be replaced with '-', and course code length over 64 will be trimmed.
2	Date and Datetime fields in this document are SG Standard Time (SST), UTC +8.